

Analisis Sentimen Ulasan Google Maps Rumah Sakit di Kota Batam

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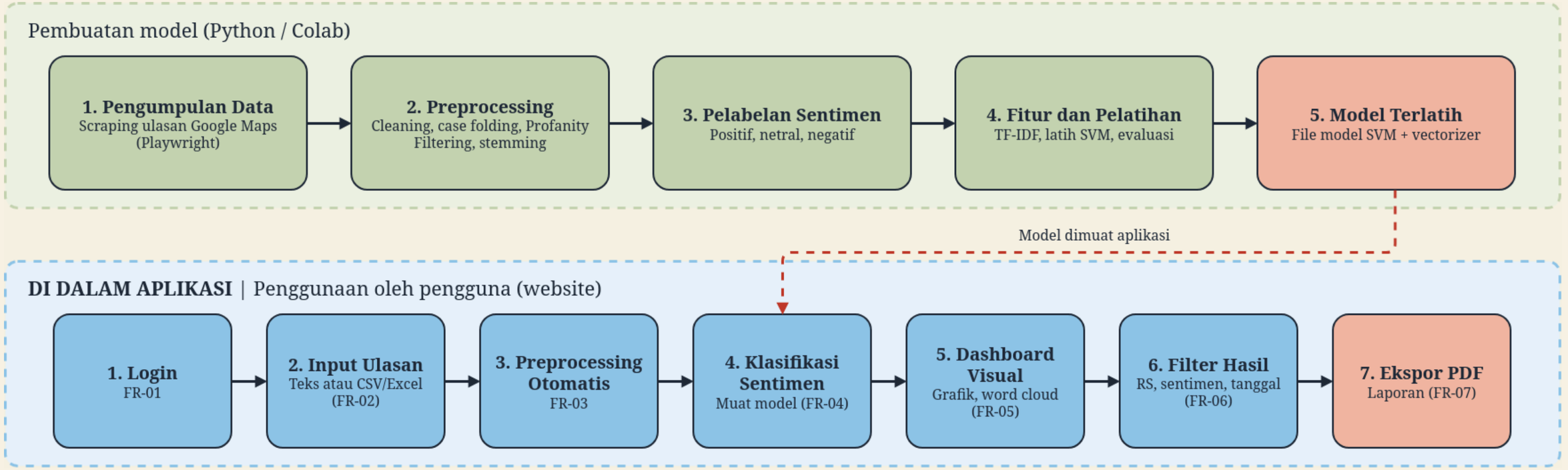
Manpro:

Agung Riyadi, S.Si., M.Kom

Background

Google Maps has become one of the primary sources of information for the public to assess the quality of hospital services through user reviews and ratings. However, the sheer volume of text-based reviews makes manual analysis time-consuming and prone to subjective judgment. Furthermore, star ratings alone are insufficient to capture users' experiences and the reasons behind their evaluations of hospital services. Therefore, this project develops a sentiment analysis system for Google Maps reviews of hospitals in Batam City using the Support Vector Machine (SVM) machine learning method. Reviews will be classified into three categories—positive, neutral, and negative—and then displayed on a visualization dashboard. This system is expected to simplify review analysis and help identify public satisfaction and complaints as input for evaluating hospital services.

System Overview



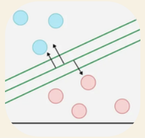
This system consists of building a model using Python/Google Colab and implementing it on a website. The process involves collecting Google Maps reviews, preprocessing, sentiment labeling, TF-IDF feature extraction, and training and evaluating an SVM model. The model is then integrated into the website so that users can enter or upload reviews, view classification results via a graphical dashboard and word cloud, filter results, and export reports in PDF format.

Technology Used

core technologies

- Python: A programming language for data processing and model development.
- Apify: Collecting Google Maps reviews via the Apify API.
- TF-IDF: Converting review text into numerical features.
- SVM (Support Vector Machine): Classifying sentiment as positive, neutral, or negative.



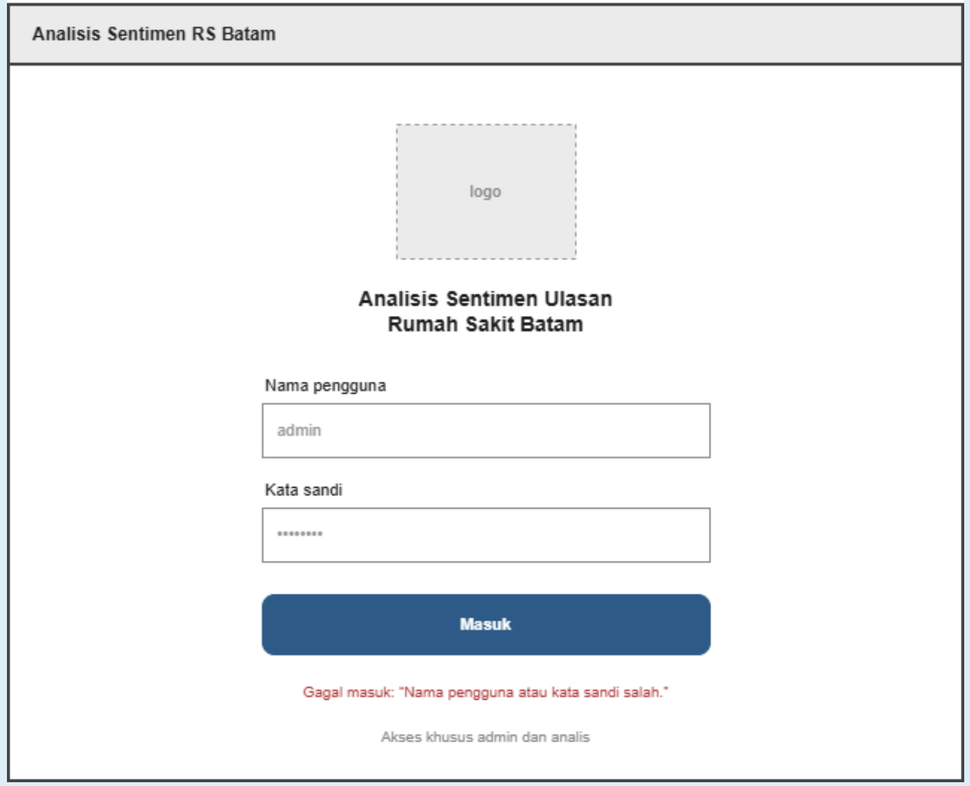


Supporting Technologies

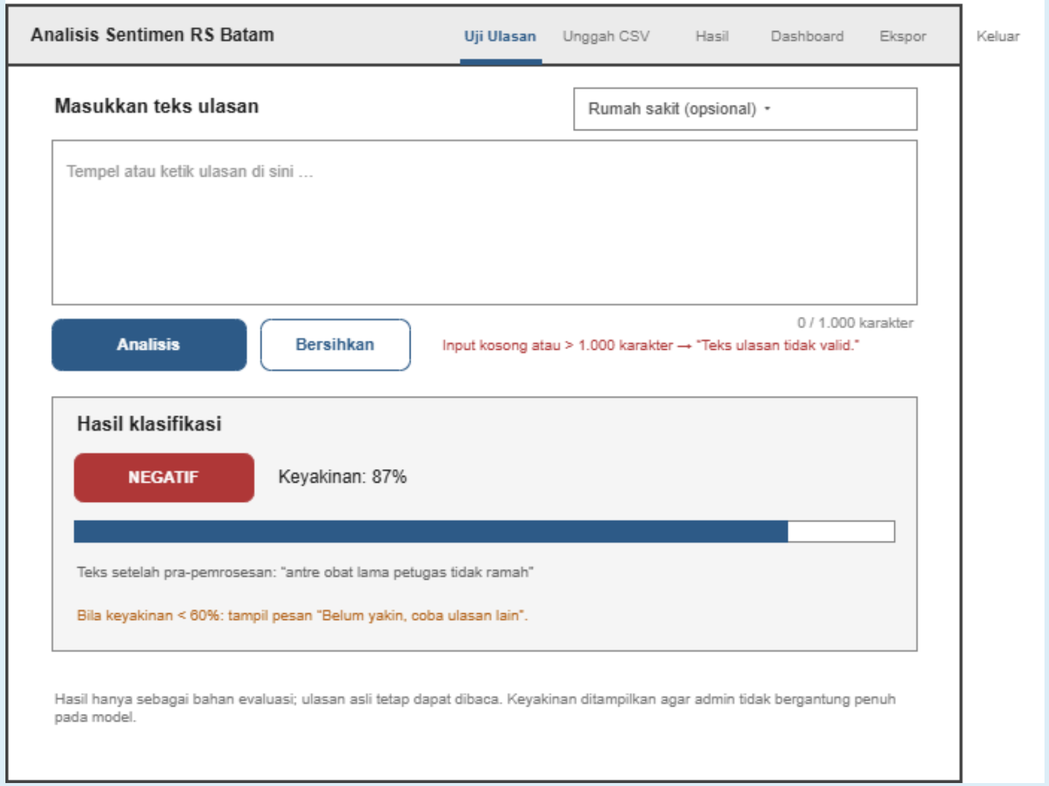
- Google Colab: An environment for model development and training.
- Pandas: Manages and processes review datasets.
- Sastrawi: Assists with Indonesian stemming.
- Scikit-learn: Supports TF-IDF, model training, and evaluation.
- HTML, CSS, JavaScript: Supports the creation of website interfaces.
- CSV/Excel & PDF: Data input formats and report export formats.

Wireframe

1. Login (FR-01)



2. Uji Klasifikasi Ulasan (US-01, FR-02, FR-04)



Wireframe

3. Unggah Data Ulasan (CSV) (FR-02, FR-03, NFR-03, NFR-05)

Analisis Sentimen RS Batam — Uji Ulasan | **Unggah CSV** | Hasil | Dashboard | Ekspor | Keluar

Unggah berkas ulasan

Seret berkas CSV/Excel ke sini atau klik untuk memilih

Kolom wajib: nama_tempat, bintang, ulasan. Berkas dengan kolom nama pengguna ditolak (privasi).

ulasan_rs_batam.csv (602 baris) — **Proses & Klasifikasi**

Memproses 340 / 602 ulasan …

Pratinjau (5 baris pertama)

nama_tempat	bintang	ulasan	label
RS Awal Bros Batam	5	Pelayanan cepat dan ramah	Positif
RSUD Embung Fatimah	1	Antre obat lama sekali	Negatif
RS Graha Hermine	3	Biasa saja	Netral
RS Budi Kemuliaan	4	Dokter ramah, parkir sempit	Positif

Berkas salah format / kosong → "Format berkas tidak sesuai. Gunakan CSV dengan kolom yang diminta."

4. Daftar Hasil Klasifikasi (FR-05, FR-06)

Analisis Sentimen RS Batam — Uji Ulasan | Unggah CSV | **Hasil** | Dashboard | Ekspor | Keluar

Rumah sakit: Semua ▾ | Sentimen: Semua ▾ | Periode: Semua ▾ | **Terapkan**

Rumah sakit	Ulasan	Label	Keyakinan
RS Awal Bros Batam	Pelayanan cepat dan ramah …	Positif	94%
RSUD Embung Fatimah	Antre obat lama sekali …	Negatif	87%
RS Graha Hermine	Biasa saja	Netral	58%
RS Budi Kemuliaan	Dokter ramah, parkir sempit	Positif	71%
RS Mutiara Aini	Petugas tidak informatif …	Negatif	90%
RS Santa Elisabeth	Dokter tanggap, pelayanan baik	Positif	96%
RS BP Batam	Ruang tunggu penuh dan lama	Negatif	82%

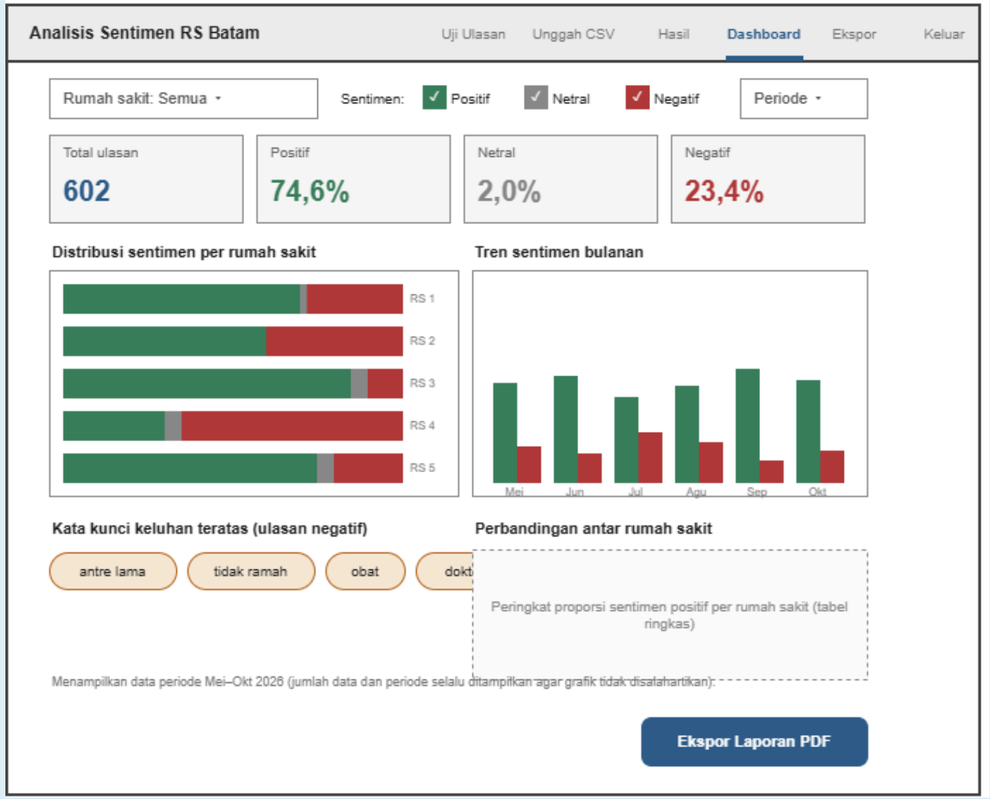
Menampilkan 1–7 dari 602 ulasan — ‹ **1** ›

Ulasan dengan keyakinan < 60% diberi tanda "Belum yakin" (baris RS Graha Hermine). Klik baris untuk membaca ulasan asli.

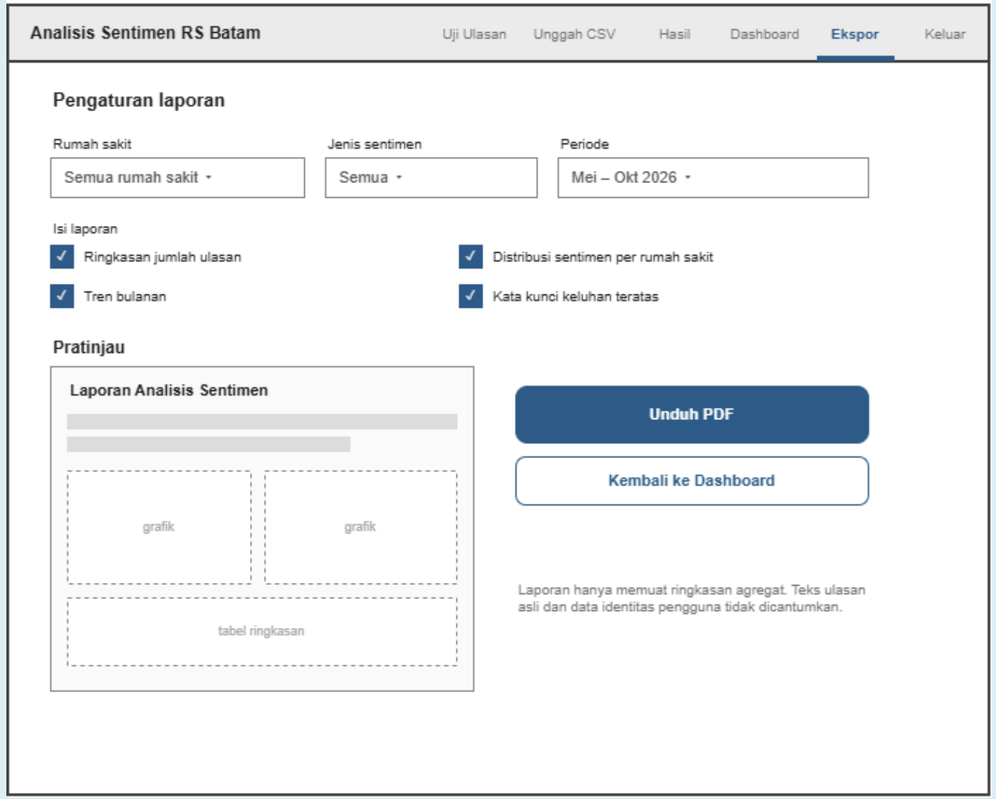
Lihat di Dashboard

Wireframe

5. Dashboard Sentimen (US-02, US-03, FR-05, FR-06)



6. Ekspor Laporan (FR-07)



Conclusion

Thank You